

ELECTRONICALLY-CONTROLLED PNEUMATIC COOLERS



STAND-ALONE SERIES VRX-100-XTRONIC

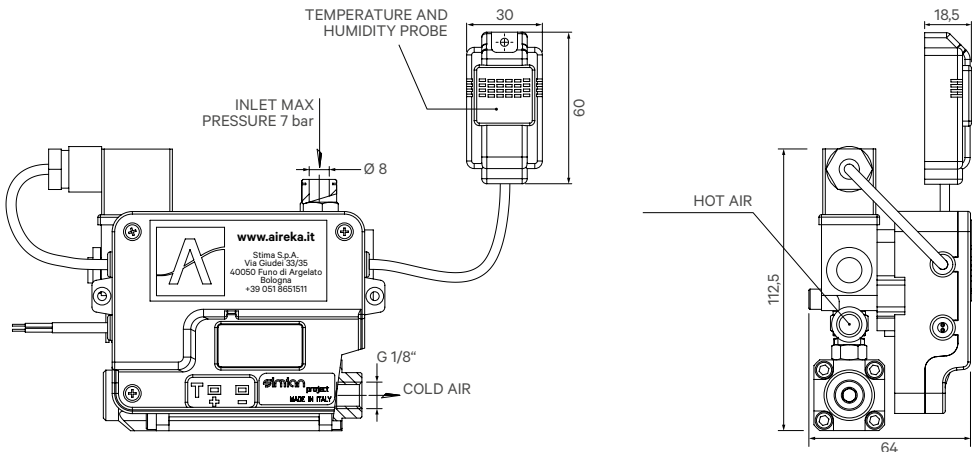
ELECTRONICALLY-CONTROLLED PNEUMATIC COOLERS



STAND-ALONE SERIES

These are pneumatic coolers with temperature-control device, in a single unit. The market increasingly demands stand-alone devices, which are able to function autonomously, based on the parameters of the ambient temperature. Therefore, we designed and developed the **XTRONIC** temperature-control units, which are available both for remote control and installed on pneumatic coolers. So, it will be enough to place one of these electronically-controlled devices inside the enclosure that has to be cooled, connect the pneumatic hoses and the electric wires, and set the desired temperature range on the display of the unit. This way, the cooler will start working only when it is necessary, it will maintain the required refrigeration, and it will make it possible to save compressed air and energy. Our **XTRONIC** control units can be customised too.

GENERAL FEATURES - VRX-100 XTRONIC	
Supply voltage	24 V DC
Recommended hose	Ø 8x1
Supply pressure	max 7 bar
Cooling power and performances	See VRX-100 (page 21)
Probe length	1 m
Temperature range	-20°C +60°C
Humidity range	00% 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil voltage and power	24 V - 3.1 W
Electric wires' section	0.75 mm
Weight	760 g

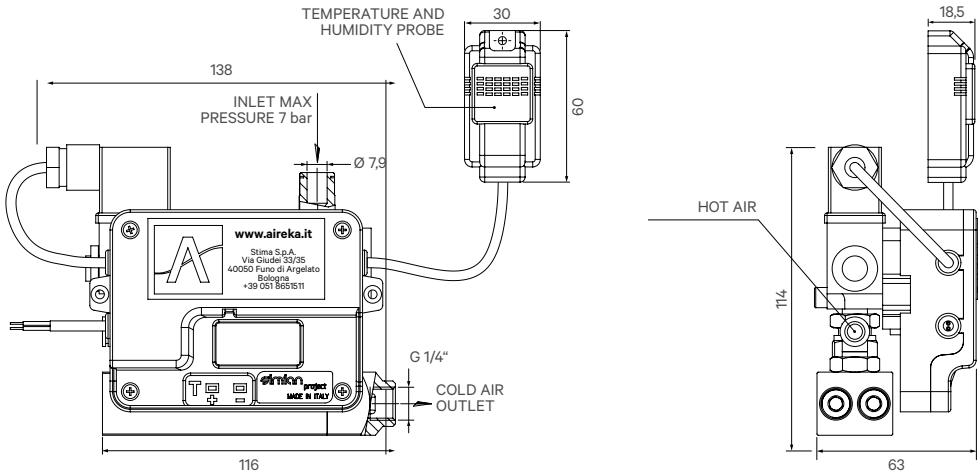


STAND-ALONE SERIES VR-200U-XTRONIC

ELECTRONICALLY-CONTROLLED PNEUMATIC COOLERS



GENERAL FEATURES - VR-200U XTRONIC	
Supply voltage	24 V DC
Recommended hose	Ø 8x1
Supply pressure	max 7 bar
Cooling power and performances	See VR-200U (page 16)
Probe length	1 m
Temperature range	-20°C +60°C
Humidity range	00% 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil voltage and power	24 V - 3.1 W
Electric wires' section	0.75 mm
Weight	770 g

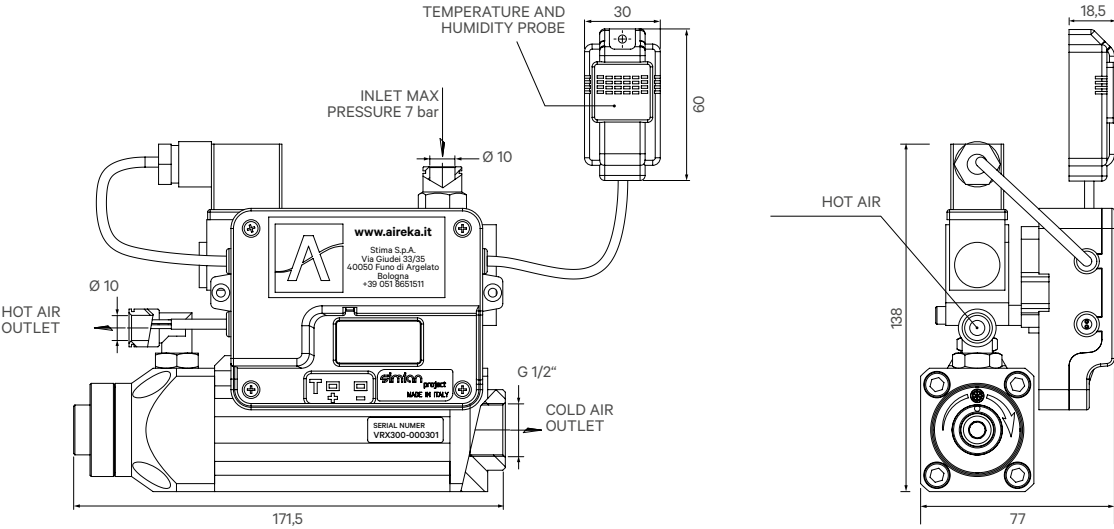


STAND-ALONE SERIES VRX-300 XTRONIC

ELECTRONICALLY-CONTROLLED PNEUMATIC COOLERS



GENERAL FEATURES - VRX-300 XTRONIC	
Supply voltage	24 V DC
Recommended hose	Ø 10x1
Supply pressure	max 7 bar
Cooling power and performances	See VRX-300 (page 22)
Probe length	1 m
Temperature range	-20°C +60°C
Humidity range	00% 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil voltage and power	24 V - 3.1W
Electric wires' section	0.75 mm
Weight	1310 g

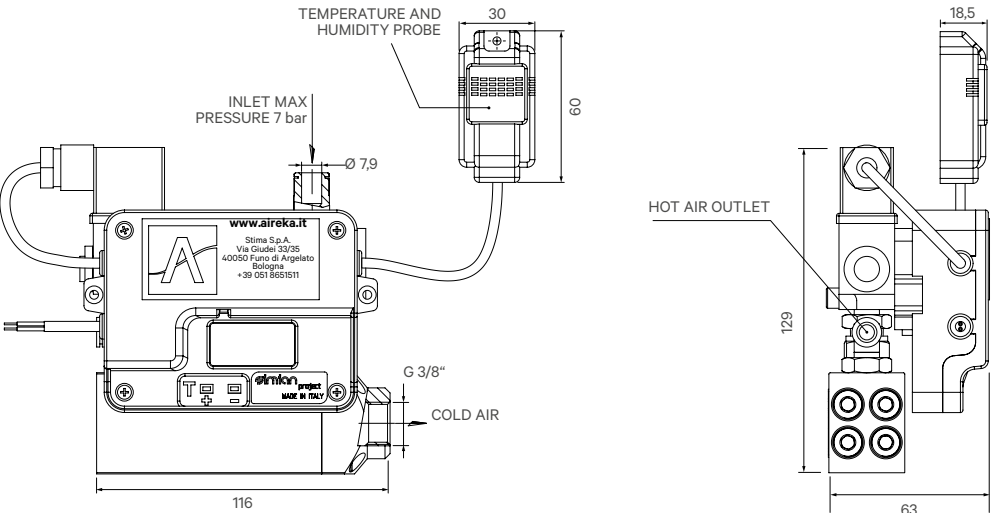


STAND-ALONE SERIES VR-400U-XTRONIC

ELECTRONICALLY-CONTROLLED PNEUMATIC COOLERS



GENERAL FEATURES - VR-400U XTRONIC	
Power supply	24 V DC
Recommended hose	Ø 8x1
Supply pressure	max 7 bar
Cooling power and performances	See VR-400U (page 16)
Probe length	1 m
Temperature range	-20°C +60°C
Humidity range	00% 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil voltage and power	24 V - 3.1 W
Electric wires' section	0.75 mm
Weight	845 g

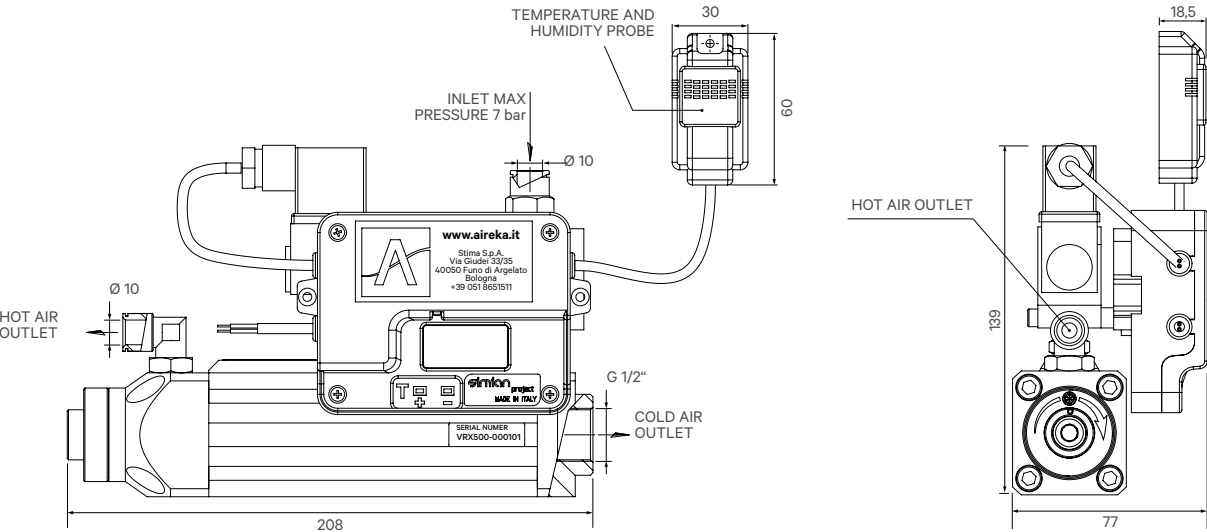


STAND-ALONE SERIES VRX-500 XTRONIC

ELECTRONICALLY-CONTROLLED PNEUMATIC COOLERS



GENERAL FEATURES - VRX-500 XTRONIC	
Supply voltage	24 V DC
Recommended hose	Ø 10x1
Supply pressure	max 7 bar
Cooling power and performances	See VR-400U (pag. 23)
Probe length	1 mt
Temperature range	-20°C +60°C
Humidity range	00% 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil voltage and power	24 V - 3.1 W
Electric wires' section	0.75 mm
Weight	1430 g



STAND-ALONE SERIES VRX-1000-XTRONIC

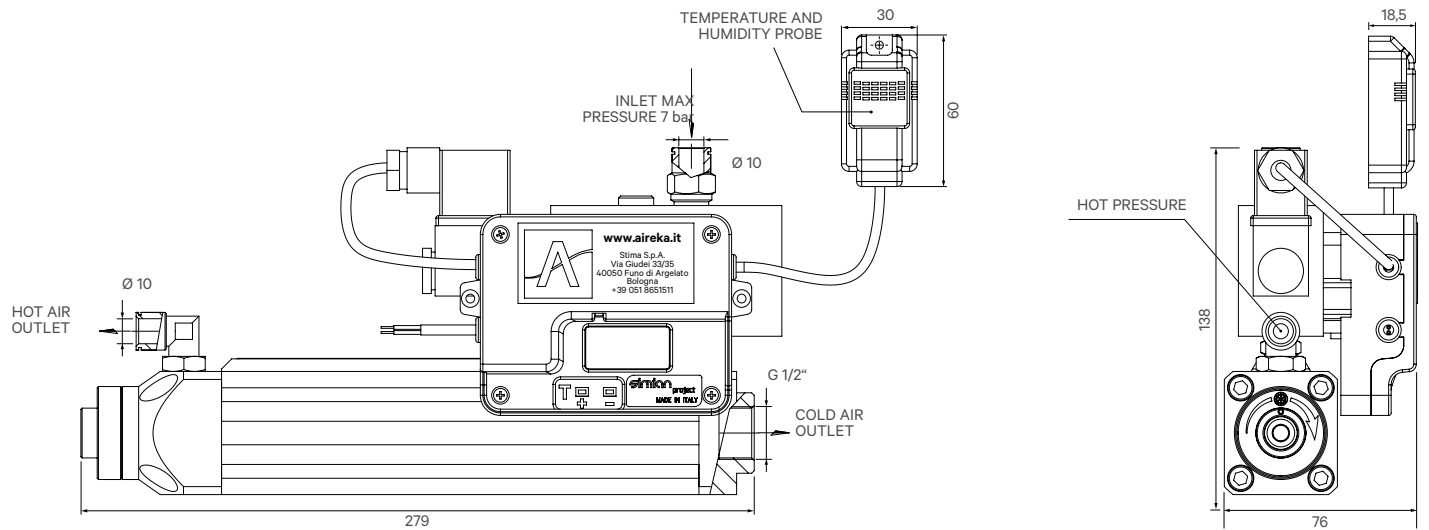
ELECTRONICALLY-CONTROLLED PNEUMATIC COOLERS

SPECIAL APPLICATIONS



GENERAL FEATURES - VRX-1000 XTRONIC

Power supply	24 V DC
Recommended hose	Ø 12x1
Supply pressure	Max 7 bar
Cooling power and performances	Vedi VRX-1000 (pag. 24)
Probe length	1 m
Temperature range	-20°C +60°C
Humidity range	00% 100% RH
Accuracy	0.1°C , 0.1 % RH
Current capacity	max 10 A
Coil voltage and power	24 V - 3.1 W
Electric wires' section	0.75 mm
Weight	1630 g



VRX300 + KDA
HEATER VERSION



VRX500 MOTION
COOLER WITH ADJUSTMENT CONTROLLABLE BY PLC



VRX300 + KDA
WITH DISTRIBUTOR OF COLD AIR



RING-COOLING SYSTEM



KDA
COLD AIR DISTRIBUTOR
ALL LENGTHS AVAILABLE ON REQUEST

COOLER AIR SAVING

PATENTED SYSTEM



The best practice in cooling down enclosures involves also a correct distribution of the cold air, after it has been produced.

To make it possible, we supply air-blowing modules, connection fittings in technopolymer, and tubes with double insulation. All this to have zero losses of cold energy and to get the highest effectiveness in your application.

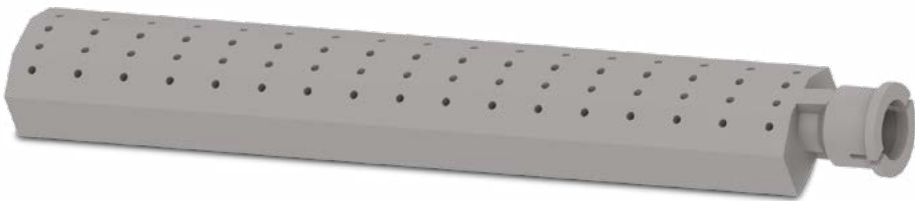
All these accessories are customisable, to meet your specific demands.



Filter against impurities, for the fixation of the hot air extractor



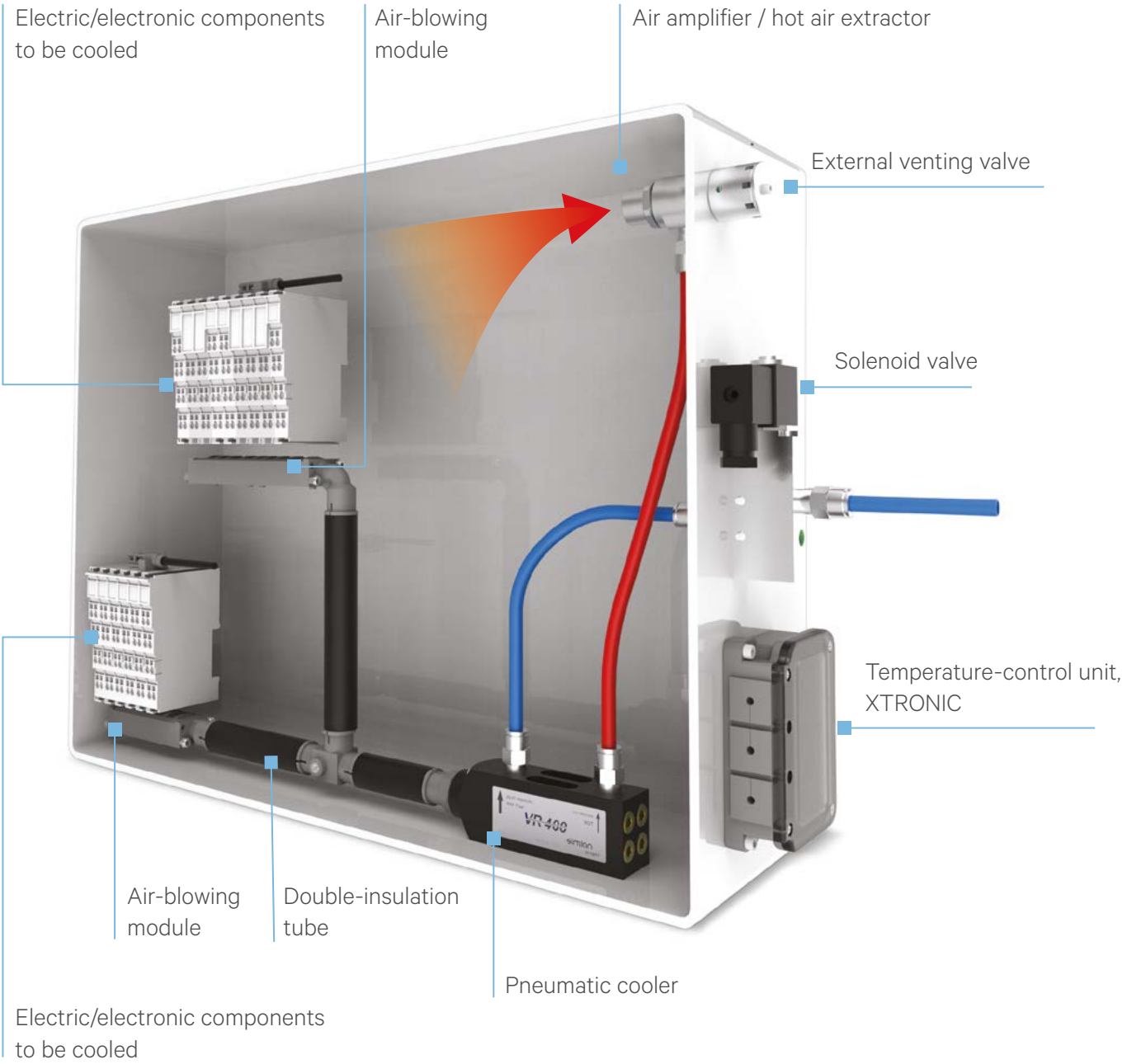
Fittings for cold air distribution



Customisable air-blowing module



Single air-blowing module



COOLER AIR SAVING

PATENTED SYSTEM



VR Series coolers and **AM Series** amplifiers used together to introduce cold air and extract hot air from electrical cabinets at the same time, using a single compressed air supply.

- Effective ventilation of the electrical cabinet
- Reduction of compressed air consumption
- Optimisation of cooling results

No matter how much cold air is introduced into an electrical cabinet, the effectiveness and efficiency of cooling will never be optimal unless the hot air generated by the electrical components is properly ventilated at the same time. With ventilation we mean both the creation of convection flows inside the cabinet which effectively distribute the air around the components, and the actual extraction of hot air from the cabinet itself.

By using the Cooler Air Saving patented system by Simian Project, two results are obtained: the first, using the VR Series coolers, is the prompt and precise cooling of the components that heat the cabinet the most. This thanks to the flexibility of installation (brackets and magnets) and the fact that the flow of cold air can be precisely directed on the main heat sources (by using adjustable nozzles). The second result is the proper ventilation of the electrical cabinet, thanks to the extraction power generated by the AM Series air amplifier, which is actuated by the hot air exhausted from the cooler.

The picture shows the system set up inside an electrical cabinet:

The VRX-500 cooler (fig.1) is actuated with compressed air from outside; the flow of cold air is directed, by using adjustable nozzles, on the electrical components that give off the most heat, while the exhaust of hot air is channelled by the red hose (fig. 3) to actuate the AM Series amplifier

The amplifier (fig.2) is mounted on the top right-hand side of the electrical cabinet; the pass-through installation allows it to suction and extract air from the cabinet; in the example of the picture, its position in the upper part of the cabinet ensures that the extraction occurs where most of the hot air accumulates and that even the electrical components located on the opposite side of the source of cold air remain at a temperature suitable for optimal functioning.

Even where pass-through mounting is not possible (for example in the event of installations in cabinets where IP protection must be guaranteed), the

fitting of the amplifier inside the cabinet ensures forced recycling of air, which eliminates the concentration of hot air in the areas located furthest away from sources of cold air.

The patented system also works well together with industrial air-conditioners in electrical cabinets with the following characteristics:

- Large electrical cabinets where the cold air generated by the air-conditioner has trouble in reaching all parts of the cabinet;
- Electrical cabinets with electrical components laid out in such a way that the convection of air around the components is tricky;
- Electrical cabinets where the heat is generated by a few components that are located far from the area where the air-conditioner introduces the cold air.

N.B.: The Cooler Air Saving system works with VRX-300, VRX-500, and VRX-1000 coolers together with AM-20ES and AM-40ES amplifiers.

